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Exploration of dimension reduction techniques for clustering spatial transcriptomics data

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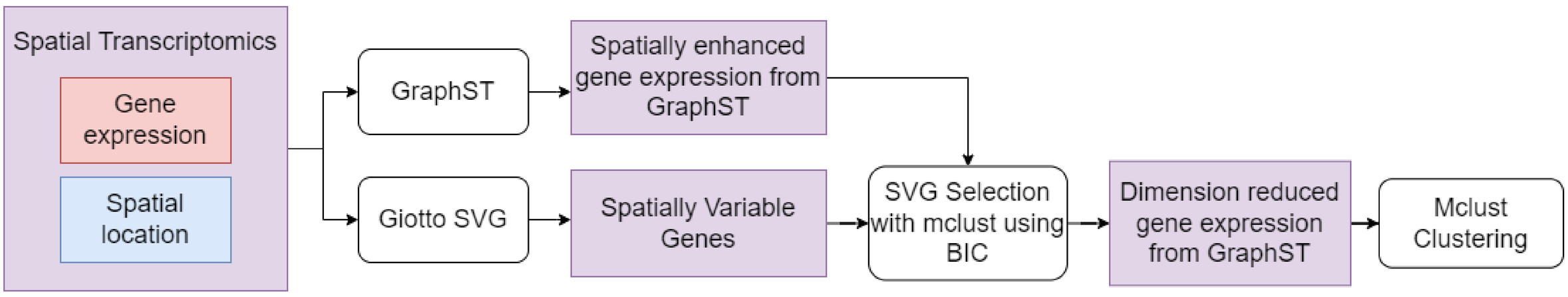
WORKFLOW



INTRODUCTION

Spatial transcriptomics (ST) provides a spatially resolved, high-dimensional assessment of gene transcription. Spatial domain identification (SDI) is a critical task in ST, as it enables a deeper understanding of tissue microenvironments and biological functions. Clustering is an essential step in SDI; however, existing methods often hardcode the number of clusters, and clustering results frequently rely on dimension reduction methods like Principal Component Analysis (PCA), which primarily captures data variability. Here, we present a spatial dimension reduction method using Giotto and mclust that eliminates the need to predefine the number of clusters or model type, achieving more promising results than PCA in the dorsolateral prefrontal cortex (DLPFC) dataset. Our work demonstrates strong potential to enhance SDI results in existing methods by improving the clustering process.

BACKGROUND

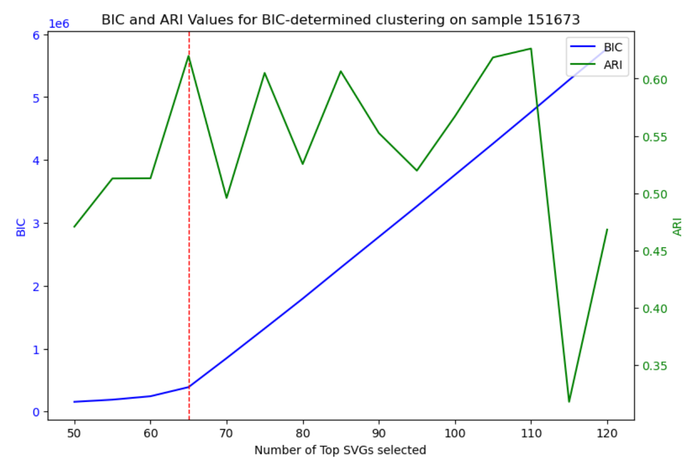
SDI typically involves a clustering step to infer spatial domains. Existing methods utilize statistical or deep learning models to incorporate spatial information for clustering. For statistical methods, Giotto implements a Hidden Markov Random Field model to detect spatial domains with consistent gene expression patterns. Among deep learning methods, GraphST uses graph convolutional networks and self-supervised contrastive learning to reconstruct gene expression matrix with spatial information.

However, existing methods face two major limitations in the clustering process. First, they often rely on a hardcoded number of clusters and/or model type. In practice, ground truth annotations, such as the number of spatial domains, are generally not available. Second, principal component analysis (PCA) is commonly used for dimension reduction of the gene expression matrix. However, PCA primarily captures variability that may not align with features needed for clustering, potentially hindering accurate domain identification.

METHOD

To overcome existing limitations, we apply model-based clustering with various dimension reduction techniques including Giotto KMeans, Giotto Rank, and Spark-X . Specifically, we optimize the clustering process in the following two aspects: First, we use the Bayesian Information Criterion (BIC) to determine the number of clusters and/or model type. The default mclust model used by GraphST is hardcoded as model EEE. By using BIC, we eliminate the need for prior knowledge in the number of clusters. Second, we apply SVG-based dimension reduction instead of PCA to reduce the dimensionality of the spatially enhanced gene expression data generated by GraphST. Specifically, we use Giotto, a statistical method, to identify genes with distinct spatial patterns, known as SVGs. We then test BIC across various numbers of top SVGs to determine the optimal number to select. The following is an example of how we use the elbow method with BIC to select the optimal SVGs for sample 151673. In this example, we select 65 as the optimal number, with an ARI of 0.6198. While this choice may not yield the highest ARI, it is adequate, as we lack ground truth information during clustering and thus cannot use ARI as an evaluation metric.

BIC and ARI for SVG Dimension Reduction



65 Top SVGs Chosen Using the Elbow Method, with an ARI of 0.6198

To directly compare the performance between GraphST's PCA and our SVG dimension reduction, we disable the spatial smoothing configuration in GraphST. This configuration calculates the mode within a spatial neighborhood to smooth cluster label assignments. For our comparison, this smoothing effect is unnecessary, as it is applied after clustering, and we focus solely on clustering performance.

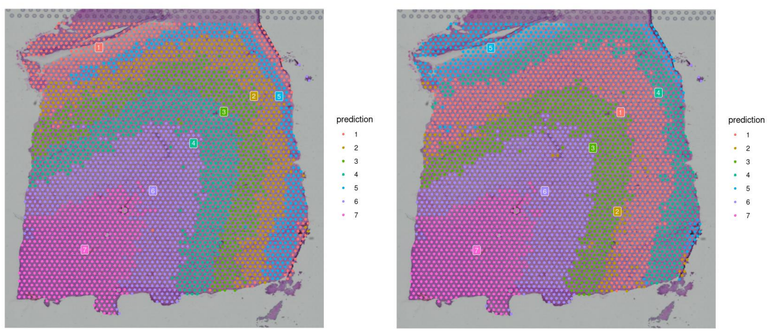
EVALUATION

We evaluate clustering performance by calculating the ARI by comparing the result clusters with the DLPFC ground truth. Our experiments are reproducible with a fixed random seed. We test a range of random seeds, and our approach consistently outperformed GraphST's PCA.

RESULTS

Using the Spatially Variable Genes (SVGs) selected by Giotto [1], we achieve a higher Adjusted Rand Index (ARI) than the PCA method used in GraphST [2] for eight out of twelve samples in the DLPFC dataset. Below is an example of the clustering results for sample 151673, where our SVG dimension reduction method yields an ARI of 0.6198, surpassing PCA's 0.5993. The left panel shows the clustering assignment of PCA, and the right panel shows the clustering assignment of our approach.

Clustering Result on Sample 151673



PCA, ARI = 0.5993

SVG Dimension Reduction, ARI = 0.6198

FUTURE WORK

We observe that the elbow method sometimes fails on certain datasets or may not select the optimal number of SVGs. Therefore, we find it necessary to develop a spatial clustering measure to determine the best number of SVGs. We are also working on generalizing our method to make it applicable across various approaches, such as embeddings generated by Variational Graph Autoencoder (VGAE) methods. Efforts are underway to ensure compatibility with these SDI methods. Additionally, further work on scalability is ongoing, as iterating through a range of top SVGs for clustering remains computationally intensive.

References: [1] Dries, R., Zhu, Q., Dong, R. et al. Giotto: a toolbox for integrative analysis and visualization of spatial expression data. Genome Biol 22, 78 (2021). https://doi.org/10.1186/s13059-021-02286-2, [2] Long, Y., Ang, K.S., Li, M. et al. Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with GraphST. Nat Commun 14, 1155 (2023). https://doi.org/10.1038/s41467-023-36796-3